

CONNOR W. MAGOON

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EDUCATION

The University of North Carolina at Chapel Hill (UNC)

PhD	Applied Mathematics	2023-2028
BS	Physics and Applied Mathematics	2019-2023

POSITIONS

Simons Foundation

Pre-Doctoral Research Intern, Biophysical Modeling Group

Jun-Aug 2026

PUBLICATIONS

Differentiation Through Black-Box Quadratic Programming Solvers <u>C. W. Magoon</u> [*] , F. Yang [*] , N. Aigerman, S. Z. Kovalsky NeurIPS Project Page	2026
The way bubbles gallop J. H. Guan, S. I. Tamim, <u>C. W. Magoon</u> , H. A. Stone, and P. J. Sáenz Physical Review Fluids	2025
Galloping Bubbles J. H. Guan [*] , S. I. Tamim [*] , <u>C. W. Magoon</u> [*] , H. A. Stone, and P. J. Sáenz Nature Communications	2025
Traveling Faraday waves J. H. Guan, <u>C. W. Magoon</u> , M. Durey, R. Camassa, and P. J. Sáenz Physical Review Fluids	2023

^{*}Equal contribution

SUBMITTED

Spontaneous galloping of a vibrating bubble along a solid boundary. Part 1: Experiments J. H. Guan, X. Liu, S. I. Tamim, <u>C. W. Magoon</u> , and P. J. Sáenz <i>Submitted</i>
Spontaneous galloping of a vibrating bubble along a solid boundary. Part 2: Simulations <u>C. W. Magoon</u> , S. I. Tamim, and P. J. Sáenz <i>Submitted</i>
Spontaneous galloping of a vibrating bubble along a solid boundary. Part 3: Theory S. I. Tamim, <u>C. W. Magoon</u> , and P. J. Sáenz <i>Submitted</i>

AWARDS

Travel Grants

NeurIPS Scholar Award 2025

Conference Awards

Gallery of Fluid Motion Award, American Physical Society / Division of Fluid Dynamics 2024
J. H. Guan, S. I. Tamim, C. W. Magoon, H. A. Stone, and P. J. Sáenz
[“Galloping Bubbles”](#)

Milton van Dyke Award, American Physical Society / Division of Fluid Dynamics 2022
J. H. Guan, C. W. Magoon, M. Durey, R. Camassa, and P. J. Sáenz
[“Run, Faraday, Run”](#)

Honors

Doctoral Merit Fellowship, UNC 2023-2024

Archibald Henderson Mathematics Medal, UNC 2023

TALKS

Collective Dynamics of Galloping Bubbles 2025
C. W. Magoon, X. Liu, S. I. Tamim, H. A. Stone and P. J. Sáenz
[APS Division of Fluid Dynamics Meeting](#)

Collective Dynamics of Galloping Bubbles 2024
C. W. Magoon, X. Liu, J. H. Guan, S. I. Tamim, and P. J. Sáenz
[APS Division of Fluid Dynamics Meeting](#)

Direct Numerical Simulations of Vibrating Bubbles Self-Propelling Along Solid Boundaries 2023
C. W. Magoon, J. H. Guan, S. I. Tamim, and P. J. Sáenz
[APS Division of Fluid Dynamics Meeting](#)

TEACHING

Instructor

Differential Equations Lab (UNC MATH 383L) Fall 2025

TA

Multivariable Calculus (UNC MATH 233) Fall 2024, 2025
General Physics (UNC PHYS 115) Summer 2021
Introductory Electromagnetism (UNC PHYS 119) Fall 2020

Grader

Fluid Mechanics (UNC MATH 563) Spring 2025
Differential Equations (UNC MATH 524) Spring 2024